

Lecture 4

One structure: canonical maps and Lagrangian submanifolds

Objectives. Course map: G1, step 4 of 9. [Lecture 2](#)'s bracket and [Lecture 3](#)'s symplectic form are one structure: $\iota_{X_f}\omega = df$ recovers the Hamiltonian vector field, and $\omega(X_f, X_g) = \{f, g\}$. Three things the next two lectures need then follow. Canonical transformations are the maps that preserve ω , and generating functions produce them. Lagrangian submanifolds, on which ω vanishes, have dimension at most n , which is why [Lecture 5](#) asks for n commuting integrals and finds Lagrangian tori. And on any cotangent bundle $p \cdot dq$ needs no coordinates, which is what the action integral of [Lecture 6](#) is made of.

4.1 The question

[Lecture 2](#) defined the Hamiltonian vector field from the Poisson bracket, and [Lecture 3](#) defined the symplectic form. The Liouville–Arnold theorem of [Lecture 5](#) needs both at once: it is about the flows of n functions in involution, and about the shape of the level set they share. So the question is

what single structure do the bracket and the forms describe — and which submanifolds does it single out?

The structure is ω itself (§ 4.2). The maps that preserve it are the canonical transformations (§ 4.3), and the submanifolds it singles out are the Lagrangian ones (§ 4.4). The worked example is the two-dimensional oscillator, whose level sets are Lagrangian tori ([Example 4.15](#)).

4.2 $\iota_{X_f}\omega = df$: one structure, two languages

[Lecture 2](#) defined X_f from the bracket, by $X_f h = \{h, f\}$ ([Definition 2.4](#)). On a symplectic manifold there is a second route, which uses no bracket at all.

Definition 4.1 (Hamiltonian vector field, from ω). On a symplectic manifold (M, ω) , the Hamiltonian vector field of a smooth function f is the unique vector field X_f with

$$\iota_{X_f}\omega = df. \tag{4.1}$$

It exists and is unique at each point by (3.10), and it is smooth because ω and f are.

Proposition 4.2 (The two definitions agree). In the coordinates of (3.11), $X_f = \sum_i (\frac{\partial f}{\partial p_i} \partial_{q_i} - \frac{\partial f}{\partial q_i} \partial_{p_i})$, which is Lecture 2's X_f exactly. Consequently

$$\omega(X_f, X_g) = \{f, g\}. \quad (4.2)$$

Proof. The first statement is the sign test (3.13), run once for each pair (q_i, p_i) with f in place of H ; the pairs do not interact, because ω does not. For the second,

$$\omega(X_f, X_g) = (\iota_{X_f} \omega)(X_g) = df(X_g) = X_g f = \{f, g\},$$

using in turn Definition 3.14, (4.1), (2.9), and Lecture 2's definition $X_g f = \{f, g\}$ (2.5). ■

On Lecture 2's worked example, $f = q^2/2$ and $g = p^2/2$ with $X_f = -q \partial_p$ and $X_g = p \partial_q$ (2.6),

$$\omega(X_f, X_g) = dq(X_f) dp(X_g) - dq(X_g) dp(X_f) = 0 \cdot 0 - p \cdot (-q) = qp = \{f, g\}.$$

Example 4.3 (Angular momentum generates rotations). Solve (4.1) a second time, for $\ell = q_1 p_2 - q_2 p_1$ on $\mathbb{R}^4$. The two sides are

$$\iota_X \omega = \sum_i (X^{q_i} dp_i - X^{p_i} dq_i), \quad d\ell = p_2 dq_1 - p_1 dq_2 - q_2 dp_1 + q_1 dp_2,$$

and matching coefficients gives

$$X_\ell = -q_2 \partial_{q_1} + q_1 \partial_{q_2} - p_2 \partial_{p_1} + p_1 \partial_{p_2} :$$

a rotation of (q_1, q_2) and, at the same rate, of (p_1, p_2) . This is the flow that Lecture 2's one-line Noether theorem was about: ℓ is conserved exactly when H is invariant under it.

Any symplectic form gives a bracket

Turn (4.2) round and make it a definition: on any symplectic manifold, put

$$\{f, g\} := \omega(X_f, X_g) = df(X_g). \quad (4.3)$$

It is bilinear and antisymmetric, and satisfies Leibniz because $d(fg) = f dg + g df$. What is left is Jacobi, and this is where closedness enters.

The Jacobi sum turns out to be $d\omega(X_f, X_g, X_h)$, a computation done in the reading below. So the bracket (4.3) satisfies the Jacobi identity exactly when $d\omega = 0$: that is what closedness in Definition 3.15 is for. With Jacobi in hand, the second proof of the engine in Lecture 2 runs unchanged, so Theorem 2.5 and Corollary 2.6 hold on every symplectic manifold.

Reading

Jacobi is $d\omega = 0$, proved.

Theorem 4.4 (Jacobi is $d\omega = 0$). Let ω be a non-degenerate 2-form, with X_f and $\{f, g\}$ defined by (4.1) and (4.3). Then for all f, g, h ,

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = d\omega(X_f, X_g, X_h). \quad (4.4)$$

Hence the bracket satisfies the Jacobi identity if and only if $d\omega = 0$.

The proof is Lecture 2's computation of the Jacobi identity with non-constant coefficients: the terms in which a derivative lands on a coefficient no longer vanish, and they assemble into exactly $d\omega$.

Proof. Work in any chart $z^1, \dots, z^{2n}$, and write $\omega_{ab} = \omega(\partial_a, \partial_b)$ and $W = (\omega_{ab})$, an antisymmetric matrix, invertible by non-degeneracy. Set $J = -W^{-1}$, also antisymmetric.

Step 1: the bracket in coordinates. Condition (4.1) reads $\sum_a X_g^a \omega_{ab} = \partial_b g$ for each b , whose solution is $X_g^a = \sum_b J^{ab} \partial_b g$. So

$$\{f, g\} = df(X_g) = \sum_{a,b} \partial_a f J^{ab} \partial_b g, \quad J^{ab} = \{z^a, z^b\}.$$

This is [Lecture 2's](#) formula (2.3) — except that J is now allowed to vary from point to point.

Step 2: what survives in the Jacobi sum. Expand the cyclic sum. The terms carrying a second derivative of one of f, g, h cancel exactly as in the proof of [Proposition 2.2](#): that argument used only that J is antisymmetric at each point, never that it is constant. What is new are the terms in which a derivative lands on J . Collecting them, with repeated indices summed,

$$\{f, \{g, h\}\} + \text{cyclic} = \partial_a f \partial_c g \partial_d h S^{acd}, \quad S^{acd} = J^{ab} \partial_b J^{cd} + J^{cb} \partial_b J^{da} + J^{db} \partial_b J^{ac}.$$

Step 3: trade ∂J for $\partial \omega$. Differentiate $JW = -\mathbb{1}$: $(\partial_b J)W + J \partial_b W = 0$, so $\partial_b J = -J(\partial_b W)W^{-1} = J(\partial_b W)J$. Substituting into each of the three terms of S and relabelling dummy indices so that every term carries the same three factors of J ,

$$S^{acd} = -J^{ab} J^{ce} J^{df} (\partial_b \omega_{ef} + \partial_e \omega_{fb} + \partial_f \omega_{be}).$$

Step 4: recognise $d\omega$. By [Definition 3.1](#), the coefficient of $dz^b \wedge dz^e \wedge dz^f$ in $d\omega$ collects exactly the three terms in the bracket, so it is the component $(d\omega)_{bef}$. And since $X_f^b = J^{ba} \partial_a f = -J^{ab} \partial_a f$, each of the three factors $\partial f J$ in $\partial_a f \partial_c g \partial_d h S^{acd}$ is minus a component of a Hamiltonian vector field. Three minus signs and the one in front of S make a plus:

$$\partial_a f \partial_c g \partial_d h S^{acd} = (d\omega)_{bef} X_f^b X_g^e X_h^f = d\omega(X_f, X_g, X_h),$$

the last step being the determinant formula (2.15) for a 3-form evaluated on three vectors. That is (4.4).

If $d\omega = 0$ the right-hand side vanishes, and Jacobi holds. Conversely, if Jacobi holds for all f, g, h , take them among the coordinate functions: the fields $X_{z^a} = J e_a$ are the columns of the invertible matrix J , so they span every tangent space, and a 3-form that vanishes on every triple from a basis is zero. ■

Example 4.5 (A non-degenerate form that is not closed). On $\mathbb{R}^4$ take $\omega' = (1 + q_2^2) dq_1 \wedge dp_1 + dq_2 \wedge dp_2$. It is non-degenerate everywhere, and not closed: $d\omega' = 2q_2 dq_2 \wedge dq_1 \wedge dp_1$. Its bracket looks perfectly reasonable — $\{q_1, p_1\} = 1/(1 + q_2^2)$, $\{q_2, p_2\} = 1$, the rest zero — and it is not a Poisson bracket:

$$\{q_1, \{p_1, p_2\}\} + \{p_1, \{p_2, q_1\}\} + \{p_2, \{q_1, p_1\}\} = \frac{2q_2}{(1 + q_2^2)^2} \neq 0.$$

[Lecture 2](#) proved the engine $X_{\{f,g\}} = -[X_f, X_g]$ from the Jacobi identity alone ([Theorem 2.5](#), second proof, a reading). So the engine runs on any closed non-degenerate 2-form, and on nothing less: that is the sense in which [Lecture 2's](#) bracket and its forms are one structure.

4.3 Canonical transformations and generating functions

Definition 4.6 (Symplectomorphism). A diffeomorphism $F: (\mathcal{M}, \omega) \rightarrow (\mathcal{M}', \omega')$ is a *symplectomorphism* if $F^*\omega' = \omega$.

In canonical coordinates, with F written as $(Q, P) = F(q, p)$, the condition is

$$\sum_i dQ_i \wedge dP_i = \sum_i dq_i \wedge dp_i, \quad (4.5)$$

where on the left Q and P are regarded as functions of (q, p) and substituted — pullback is substitution (§ 3.4). It is a condition you already know under another name, since the matrix of the pulled-back form is $D^T \Omega D$, with D the Jacobian of F .

Proposition 4.7 (Symplectomorphisms are the canonical transformations). $(Q, P) = F(q, p)$ satisfies (4.5) if and only if the new coordinates have canonical brackets, $\{Q_i, P_j\} = \delta_{ij}$ and $\{Q_i, Q_j\} = \{P_i, P_j\} = 0$.

Proof (★). By (3.12) the pulled-back form has matrix $D^T \Omega D$, so (4.5) says $D^T \Omega D = \Omega$. On the other hand, by the chain rule and $\{z^a, z^b\} = \Omega^{ab}$, the brackets of the new coordinates form the matrix $D \Omega D^T$, and canonical brackets say $D \Omega D^T = \Omega$. The two conditions are equivalent: either one forces $\det D = \pm 1$, and inverting $D^T \Omega D = \Omega$ gives $D^{-1} \Omega^{-1} D^{-T} = \Omega^{-1}$, which with $\Omega^{-1} = -\Omega$ is $D \Omega D^T = \Omega$; the converse is the same computation backwards. ■

For one degree of freedom $dQ \wedge dP = (\det D) dq \wedge dp$, so a *canonical transformation of the plane is exactly an area- and orientation-preserving map*. To check the shear $(q, p) \mapsto (q + p^2, p)$, pull back: $d(q + p^2) \wedge dp = dq \wedge dp + 2p dp \wedge dp = dq \wedge dp$. The stretch $(q, p) \mapsto (2q, p)$ fails, with $\{2q, p\} = 2$. For $n \geq 2$ the condition is much stronger than preserving volume (a reading at the end of this lecture).

Reading

Proposition 4.8 (Hamiltonian flows are symplectic · proved on $\mathbb{R}^{2n}$; modulo Darboux in general). For every t and every point at which the flow φ_t of X_H is defined, $\varphi_t^* \omega = \omega$.

Proof. First on $\mathbb{R}^{2n}$ with the standard form. By Proposition 4.2, $X_H = \Omega \nabla H$ in the (q, p) ordering. The Jacobian $D(t)$ of φ_t at a point therefore obeys the linearised equation $\dot{D} = A(t)D$, $D(0) = \mathbb{1}$, with $A = \Omega S$ and S the (symmetric) Hessian of H along the trajectory. Then

$$\frac{d}{dt}(D^T \Omega D) = D^T (A^T \Omega + \Omega A) D = D^T (S \Omega^T \Omega + \Omega^2 S) D = D^T (S - S) D = 0,$$

because $\Omega^T \Omega = \mathbb{1}$ and $\Omega^2 = -\mathbb{1}$. So $D^T \Omega D = \Omega$ for all t , which is the claim on $\mathbb{R}^{2n}$; the reduction of a general symplectic manifold to this case is in the reading. ■

On a general symplectic manifold the trajectory may leave any one chart, but $\varphi_{t+s} = \varphi_s \circ \varphi_t$ gives

$$\frac{d}{dt} \varphi_t^* \omega = \varphi_t^* \left(\frac{d}{ds} \Big|_{s=0} \varphi_s^* \omega \right),$$

and the bracket is a rate of change at a single point, which may be computed in a Darboux chart there (Theorem 4.18): it is the computation above at $t = 0$, where $D = \mathbb{1}$, and it vanishes.

With canonical brackets, Leibniz and the chain rule give $\dot{Q}_i = \{Q_i, H\} = \partial H / \partial p_i$ and $\dot{P}_i = -\partial H / \partial Q_i$: Hamilton's equations keep their form.

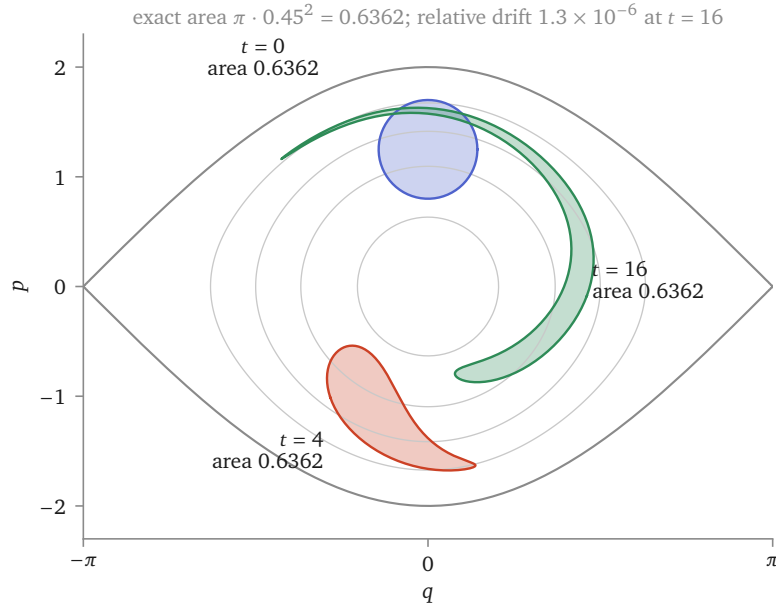


Figure 4.1: A symplectomorphism distorting a region and preserving its area: a disc of librating pendulum states carried by the flow for times 0, 4, 16. Outer orbits are slower (Lecture 6); that Hamiltonian flows are symplectic is proved in the reading above, so the disc shears into a filament. The areas are computed from the transported boundaries; grey is the separatrix.

Generating functions

A transformation given as a formula is easy to check. The problem in practice runs the other way: to *produce* canonical transformations adapted to a given H . Generating functions do this, and the form language says in one line why they work.

Proposition 4.9 (Why generating functions work). *Let $S(q, P)$ be a function of n old positions and n new momenta, and define $(q, p) \mapsto (Q, P)$ implicitly by*

$$p_i = \frac{\partial S}{\partial q_i}, \quad Q_i = \frac{\partial S}{\partial P_i}, \quad (4.6)$$

wherever (q, P) are good coordinates. Then (4.5) holds.

Proof. In the coordinates (q, P) , (4.6) says exactly that

$$\sum_i p_i dq_i + \sum_i Q_i dP_i = dS.$$

Apply d . The right-hand side gives $ddS = 0$ (Theorem 3.2); the left gives $\sum dp_i \wedge dq_i + \sum dQ_i \wedge dP_i$. So $\sum dQ_i \wedge dP_i = -\sum dp_i \wedge dq_i = \sum dq_i \wedge dp_i$. ■

That is the mechanism in full: *a generating function is a primitive for a combination of canonical 1-forms, and $d \circ d = 0$ turns “exact” into “canonical”*. For example $S = \sum_i q_i P_i + f(q)$ gives $Q = q$ and $P = p - \nabla f$: shifting every momentum by a gradient is canonical. The function $\sum_i q_i P_i$ alone gives the identity.

Reading

The other classical types work identically. A *type-1* function $S(q, Q)$ with $p = \partial S / \partial q$ and $P = -\partial S / \partial Q$ says $p \cdot dq - P \cdot dQ = dS$; for instance $S = qQ$ gives $(Q, P) = (p, -q)$, which exchanges position and momentum up to a sign. Type 1 cannot generate the identity, whose q and Q are not independent, which is one reason type 2 is the workhorse and the type [Lecture 6](#) uses.

Hamilton–Jacobi, recalled in this language

Hamilton–Jacobi looks for a type-2 generating function adapted to H : an $S(q, P)$ with

$$H\left(q, \frac{\partial S}{\partial q}\right) = K(P) \quad (4.7)$$

for some function K of the new momenta alone. The new equations are then $\dot{P}_i = -\partial K / \partial Q_i = 0$ and $\dot{Q}_i = \partial K / \partial P_i$, a constant: conserved momenta, and positions advancing at constant speed. [Lecture 6](#) does this with $P = I$ the actions and $Q = \varphi$ the angles. With P held fixed, (4.7) also says that the n -dimensional set $\{(q, \partial S / \partial q)\}$ lies inside a level set of H — [§ 4.4](#) returns to this.

4.4 Lagrangian submanifolds

How large can a subspace be on which ω vanishes?

Let $\dim V = 2n$ with ω non-degenerate. The ω -complement of a subspace U is $U^\omega = \{v : \omega(v, u) = 0 \text{ for all } u \in U\}$; by (3.10) it is the preimage of the covectors vanishing on U , so

$$\dim U^\omega = 2n - \dim U. \quad (4.8)$$

Unlike an orthogonal complement, U^ω can contain U , since every vector is ω -orthogonal to itself. Call U *isotropic* if ω vanishes on it, i.e. $U \subseteq U^\omega$; then (4.8) gives $\dim U \leq 2n - \dim U$: *an isotropic subspace has dimension at most n .*

Definition 4.10 (Lagrangian subspace, Lagrangian submanifold). A *Lagrangian subspace* is an isotropic subspace of the maximal dimension n . A submanifold $L \subset \mathcal{M}$ is *Lagrangian* if it has dimension n and $\omega(v, w) = 0$ for all v, w tangent to L — equivalently, if the pullback of ω to L vanishes, written $\omega|_L = 0$.

This bound is why [Lecture 5](#) asks for exactly n integrals.

Proposition 4.11 (At most n commuting integrals). If $F_1, \dots, F_k$ satisfy $\{F_i, F_j\} = 0$ and $dF_1, \dots, dF_k$ are linearly independent at a point, then $k \leq n$.

Proof. At that point the vectors X_{F_i} are linearly independent, since $\iota_{X_{F_i}} \omega = dF_i$ and $v \mapsto \iota_v \omega$ is an isomorphism. They span an isotropic subspace, since $\omega(X_{F_i}, X_{F_j}) = \{F_i, F_j\} = 0$ by (4.2). So $k \leq n$. ■

Seeing it in $\mathbb{R}^4$

In $\mathbb{R}^4$, $\omega(v, w) = A_1 + A_2$, the signed areas of the two shadows of the parallelogram on v, w . A plane is Lagrangian when the two shadow areas of a basis parallelogram *cancel* ([Figure 4.2](#)).

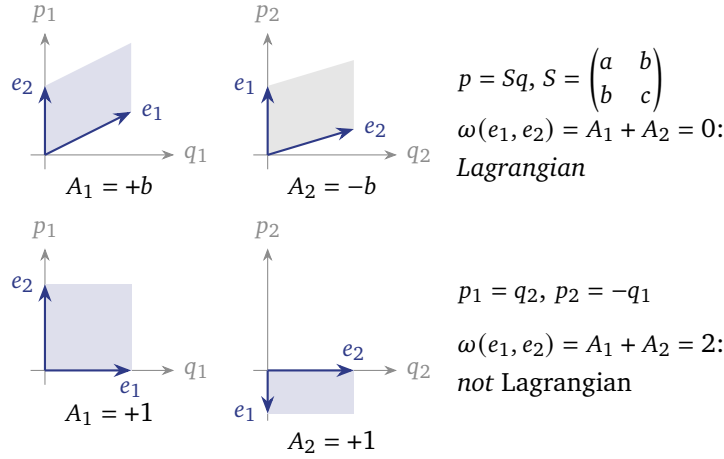


Figure 4.2: Two planes in $\mathbb{R}^4$, each drawn through the shadows of a basis in the two symplectic planes; coordinates are ordered (q_1, q_2, p_1, p_2) . Top: $e_1 = (1, 0, a, b)$, $e_2 = (0, 1, b, c)$ span the graph of a symmetric matrix (drawn with $a = 0.5$, $b = 0.8$, $c = 0.3$); the shadow areas are $+b$ and $-b$ and cancel. Bottom: $e_1 = (1, 0, 0, -1)$, $e_2 = (0, 1, 1, 0)$ span the graph of an antisymmetric one; both shadows run anticlockwise and the areas add. ω only ever looks at these two shadows.

Examples

Example 4.12 (Curves in the plane). For $n = 1$ every curve is Lagrangian: it has dimension $1 = n$, and a 2-form vanishes on parallel vectors. The notion has content from $n = 2$ on.

Example 4.13 (Positions and momenta). In $\mathbb{R}^{2n}$ the q -space $\{p = 0\}$ and each p -space $\{q = \text{const}\}$ are Lagrangian, since ω pairs a q -direction only with a p -direction. The (q_1, p_1) -plane is not: ω restricts to $dq_1 \wedge dp_1$ on it.

The most important examples are graphs, and they bring back [Lecture 3](#).

Proposition 4.14 (Lagrangian graphs are closed 1-forms). Let $\sigma = \sum_i \sigma_i(q) dq_i$ be a 1-form on an open set of q -space, and $L_\sigma = \{(q, p) : p_i = \sigma_i(q)\}$ its graph. Then $\omega|_{L_\sigma} = -d\sigma$. Hence L_σ is Lagrangian if and only if σ is closed. In particular the graph of dS ,

$$p_i = \frac{\partial S}{\partial q_i}, \quad (4.9)$$

is Lagrangian for every function S .

Proof. Parametrise L_σ by q . Pulling back is substituting $p_i = \sigma_i(q)$:

$$\omega|_{L_\sigma} = \sum_i dq_i \wedge d\sigma_i = \sum_{i,j} \frac{\partial \sigma_i}{\partial q_j} dq_i \wedge dq_j = - \sum_{i,j} \frac{\partial \sigma_j}{\partial q_i} dq_j \wedge dq_i = -d\sigma,$$

by (2.13) and (3.1). For $\sigma = dS$ this vanishes by [Theorem 3.2](#). ■

S is called a *generating function* of the Lagrangian submanifold, and the name is not a coincidence: by (4.7), a solution of Hamilton–Jacobi is a family of Lagrangian graphs, one for each value of P , each lying in a level set of H . [Lecture 6](#)’s $S(q, I)$ is exactly that, with the invariant tori as the family.

Reading

By the Poincaré lemma a closed σ is locally some dS ; globally it need not be. On the pendulum's cylinder, with angle q , a rotation orbit at energy $E > 1$ is the graph $p = \sqrt{2(E + \cos q)}$ of the 1-form $\sigma = \sqrt{2(E + \cos q)} dq$ — closed, as every 1-form on a circle is, and Lagrangian. It is not dS for any function S on the circle, for the reason of [Proposition 3.6](#): $\oint \sigma > 0$. [Lecture 6](#) calls $\frac{1}{2\pi} \oint \sigma$ the rotating pendulum's action. A nonzero action is precisely the obstruction to a global generating function, and [Lecture 6](#) lives in that gap.

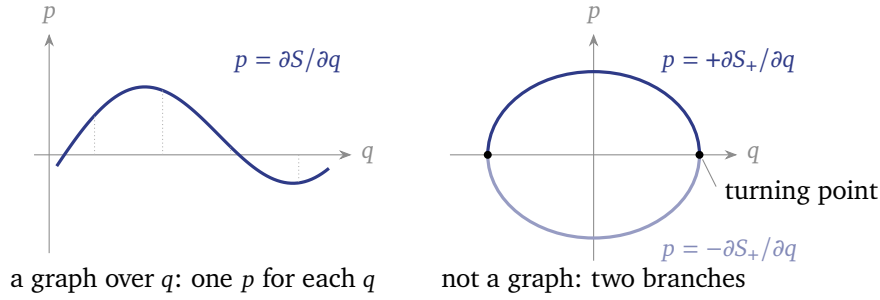


Figure 4.3: Left: the graph of dS , Lagrangian for every S ([Proposition 4.14](#)). Right: an oscillator orbit, Lagrangian by [Example 4.12](#) but not a graph — two points over each q , and a vertical tangent at the turning points. Each branch is a graph, generated by $\pm S_+$; the turning points are where [Lecture 6](#)'s $S(q, I)$ changes branch.

Example 4.15 (A Lagrangian torus in $\mathbb{R}^4$). Take the two-dimensional oscillator's integrals $F_i = \frac{1}{2}(q_i^2 + p_i^2)$ and, for $c_1, c_2 > 0$, the level set

$$T = \{F_1 = c_1, F_2 = c_2\} = \{q_1^2 + p_1^2 = 2c_1\} \times \{q_2^2 + p_2^2 = 2c_2\},$$

a circle in each symplectic plane: a torus ([Figure 4.4](#)). Its tangent spaces are spanned by $X_{F_1} = p_1 \partial_{q_1} - q_1 \partial_{p_1}$ and $X_{F_2} = p_2 \partial_{q_2} - q_2 \partial_{p_2}$, each tangent to its own circle and nonzero there, and $\omega(X_{F_1}, X_{F_2}) = \{F_1, F_2\} = 0$. So T is Lagrangian: each tangent parallelogram has one edge in each symplectic plane, so both its shadows are degenerate.

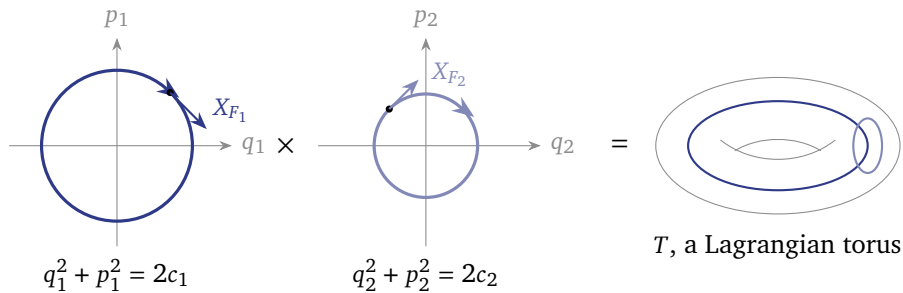


Figure 4.4: A level set of the 2D oscillator: a product of two circles, one in each symplectic plane, with X_{F_1}, X_{F_2} tangent and ω -orthogonal. So it is Lagrangian. [Lecture 5](#) proves that this is no accident: each compact connected component of a regular level set of n independent commuting integrals is a Lagrangian torus.

[Example 4.15](#) is [Lecture 5](#)'s first proof step, done by hand; the general argument is [Proposition 4.11](#)'s, with equality. Like the circle of [Figure 4.3](#), T is not a graph over (q_1, q_2) .

4.5 Cotangent bundles and the canonical 1-form

Positions and their momenta

Let Q be a configuration space, a manifold of dimension n — the circle for the pendulum, the torus T^2 for the double pendulum. Its *cotangent bundle* is the set of pairs

$$T^*Q = \{(q, p) : q \in Q, p \in T_q^*Q\}, \quad \pi(q, p) = q,$$

a point of configuration space and a covector there. A chart $q_1, \dots, q_n$ on Q writes the covector as $p = \sum_i p_i dq_i$ (Definition 2.7), and (q, p) is then a chart on the $2n$ -dimensional T^*Q . That momenta are covectors shows in a change of chart $q' = \phi(q)$: writing the same covector in both bases, $\sum_j p'_j dq'_j = \sum_i p_i dq_i$, forces

$$p_i = \sum_j p'_j \frac{\partial q'_j}{\partial q_i} : \quad (4.10)$$

momenta transform like the components of a gradient, not like those of a velocity. For polar coordinates on the plane, (4.10) gives $p_r = (x p_x + y p_y)/r$ and $p_\theta = x p_y - y p_x$ — the momentum conjugate to the angle is the angular momentum, as it should be.

The canonical 1-form

Definition 4.16 (Canonical 1-form). At a point (q, p) of T^*Q and for a tangent vector v there,

$$(p \cdot dq)_{(q,p)}(v) = p(d\pi v). \quad (4.11)$$

In words: push v down to configuration space, and feed it to the covector p — which is part of the point, and so is already there (Figure 4.5).

In a chart, $d\pi v = \sum_i v^{q_i} \partial_{q_i}$ drops the momentum components of v , so (4.11) is $(\sum_i p_i dq_i)(v)$. The form of Example 2.9, and of Lecture 6's action integral, is therefore *the same 1-form in every cotangent chart*, because (4.11) mentions no chart.

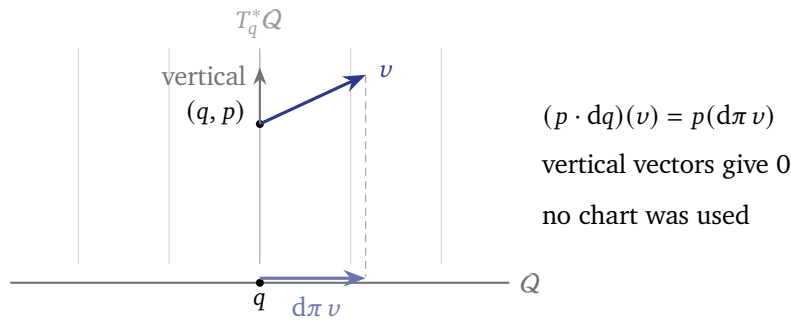


Figure 4.5: The canonical 1-form. Above each q sits the fibre T_q^*Q of momenta there. A vector v at (q, p) is pushed down to $d\pi v$ and fed to p , which is part of the point. A vector along the fibre projects to zero and is not seen.

Theorem 4.17 (Every cotangent bundle is symplectic). $\omega = -d(p \cdot dq)$ is a symplectic form on T^*Q , and in every cotangent chart it is $\sum_i dq_i \wedge dp_i$.

Proof. In a chart, $d(\sum_i p_i dq_i) = \sum_i dp_i \wedge dq_i = -\sum_i dq_i \wedge dp_i$, so ω is the standard form there, which is non-degenerate (Example 3.16). It is closed because it is exact: $d\omega = -dd(p \cdot dq) = 0$. And it is the same form in every chart, because $p \cdot dq$ is. ■

So the pendulum's cylinder, the double pendulum's T^*T^2 and the rigid body's $T^*SO(3)$ are symplectic, with no choice made.

The minus sign, and why it costs nothing

The minus sign in $\omega = -d(p \cdot dq)$ comes with the wedge order $dq \wedge dp$, which (3.13) showed is the one giving Hamilton's equations. It is compensated where it matters. Orbits of the oscillator run clockwise in the (q, p) -plane, a clockwise orbit bounds its disc with the opposite orientation, and the two signs cancel: $\oint p dq$ along the motion is $+\int_D \omega$, the enclosed area, which is positive (§ 3.4 computed it). So the action integral of Lecture 6, taken along the motion, comes out positive.

Reading

Point transformations are canonical

A diffeomorphism ϕ of configuration space moves momenta by the transpose of its inverse Jacobian — that is (4.10) read as a map — and so lifts to a map $\hat{\phi}(q, p) = (\phi(q), p')$ of T^*Q , the *cotangent lift*. Because $p \cdot dq$ is defined without a chart, $\hat{\phi}^*(p \cdot dq) = p' \cdot dq$, and hence $\hat{\phi}^*\omega = \omega$ by (3.6). Every change of configuration coordinates, with the momenta carried along as covectors, is a canonical transformation. This is why passing to polar coordinates, with p_θ the angular momentum, never needs checking. It is also the type-2 generating function $S(q, p') = \sum_j p'_j \phi_j(q)$ of Proposition 4.9, as a one-line computation with (4.6) confirms.

4.6 What it buys

- *The count in the definition of integrability.* At most n independent functions Poisson-commute (Proposition 4.11), because their Hamiltonian vector fields span a subspace on which ω vanishes. So n is the most one can ask for, and Lecture 5 asks for exactly that.
- *The shape of the invariant tori.* The oscillator's level sets are Lagrangian tori (Example 4.15). Lecture 5 proves the same of every compact connected component of a common level set of n functions in involution on which their differentials are independent.
- *The tool for action–angle variables.* A type-2 generating function $S(q, P)$ produces a canonical transformation (Proposition 4.9), and a solution of Hamilton–Jacobi is a family of Lagrangian graphs. Lecture 6 builds the action–angle variables from such an $S(q, I)$.
- *A chart-free $p \cdot dq$.* On any cotangent bundle the canonical 1-form needs no coordinates (Definition 4.16), and $\omega = -d(p \cdot dq)$ (Theorem 4.17), so the action integral of Lecture 6 is the same in every chart.

4.7 Caveats, in one place

- *Status.* Of the results the lectured core uses, three are proved only in readings: the Jacobi identity for the bracket of ω , that symplectomorphisms are the maps with canonical brackets (Proposition 4.7), and that Hamiltonian flows are symplectic. So the engine and Corollary 2.6 on a general symplectic manifold rest on two reading proofs, this lecture's and the second proof of the engine in Lecture 2. Darboux's theorem is sketched, and its version adapted to commuting functions is cited, both in a reading. Propositions 4.2, 4.9, 4.11 and 4.14 and Theorem 4.17 are proved in the lectured core.
- *Generating functions are local.* Proposition 4.9 holds where (q, P) are good coordinates. A closed 1-form need not have a global primitive, and Lecture 6's $S(q, I)$ changes branch at the turning points (Figure 4.3).

Reading

4.8 Darboux's theorem: no local invariants

Theorem 4.18 (Darboux · sketched). *Every point of a $2n$ -dimensional symplectic manifold (M, ω) has a chart $q_1, \dots, q_n, p_1, \dots, p_n$ around it in which $\omega = \sum_i dq_i \wedge dp_i$. (da Silva, §8.1; Arnold, §43.)*

Such a chart is a *Darboux chart*, or *canonical coordinates*. So any statement about ω alone — or about the bracket and the fields X_f built from it — that is proved in q 's and p 's holds near every point of every symplectic manifold.

What “no local invariants” means — and what it does not

Compare a Riemannian metric, which measures lengths and angles. A sphere of radius R has curvature $1/R^2$ at every point, and curvature is a *local invariant*: no chart, however small, makes the sphere's metric into the flat $dx^2 + dy^2$. Darboux's theorem says a symplectic form has nothing of the kind (Figure 4.6): every small patch of every $2n$ -dimensional symplectic manifold is indistinguishable from every other, and the dimension is the only local invariant there is.

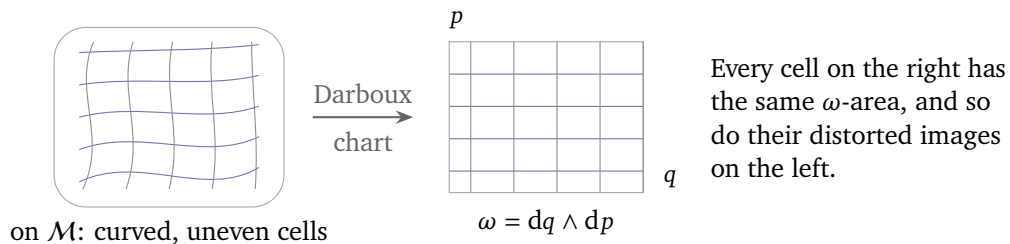


Figure 4.6: A Darboux chart straightens a symplectic form. Contrast a curved surface, where no chart can make the metric flat even on a tiny patch.

It is easy to hear “no local invariants” as “nothing interesting happens”, and that is the opposite of the point: everything interesting is *global*. It lives in loops that do not bound, in regions of finite size, in how charts are glued together — never at a point. The actions of

Lecture 6 are integrals over loops, and the tori of Lecture 5 are compact; neither is visible in a Darboux chart.

Why it is true: Moser's idea

At a single point, linear algebra already makes ω standard (the reading proves this). Call that constant form ω_0 and join it to ω by the straight line $\omega_t = \omega_0 + t(\omega - \omega_0)$, $0 \leq t \leq 1$. Near the point each ω_t is non-degenerate, and $\omega - \omega_0$ is closed, hence $\omega - \omega_0 = d\sigma$ by the Poincaré lemma (Theorem 3.4). Now look for a flow ψ_t that pulls every ω_t back to ω_0 . Its velocity V_t is forced: $\iota_{V_t}\omega_t = -\sigma$, which non-degeneracy solves. That this V_t does the job is one differentiation, which uses $d\omega_t = 0$ and a formula for differentiating a form along a flow that we cite (reading). Then ψ_1 carries ω to ω_0 , and its coordinates are a Darboux chart.

At a point it is linear algebra; near a point it is Moser

Lemma 4.19 (Linear Darboux). *Let ω be a non-degenerate antisymmetric bilinear form on a vector space V . Then V has a basis $e_1, \dots, e_n, f_1, \dots, f_n$ with $\omega(e_i, f_j) = \delta_{ij}$ and $\omega(e_i, e_j) = \omega(f_i, f_j) = 0$. In particular $\dim V = 2n$.*

Proof. Pick any $e_1 \neq 0$. By non-degeneracy some f_1 has $\omega(e_1, f_1) \neq 0$; rescale it to 1. On $U = \text{span}(e_1, f_1)$ the form is non-degenerate, and its ω -complement $U^\omega = \{v : \omega(v, e_1) = \omega(v, f_1) = 0\}$ has dimension $\dim V - 2$ and meets U only in 0. So $V = U \oplus U^\omega$, and ω is still non-degenerate on U^ω : a vector there pairing to zero with all of U^ω also pairs to zero with U , hence with V . Repeat inside U^ω . ■

Lemma 4.19 makes ω standard at one point by a linear change of coordinates. A metric can be pushed a little further — suitable coordinates make it Euclidean at a point to first order — but no further: at second order its curvature is in the way. A symplectic form can be made standard on a whole neighbourhood, and that is Theorem 4.18. The proof spends closedness in exactly one place.

Proof of Theorem 4.18 (sketch: Moser's trick). Work in a chart around a point z_0 in which, by Lemma 4.19, $\omega_1 := \omega$ agrees at z_0 with the constant form $\omega_0 = \sum dq_i \wedge dp_i$. Join them by $\omega_t = \omega_0 + t(\omega_1 - \omega_0)$, $t \in [0, 1]$. Each ω_t is closed, and equals ω_0 at z_0 , so it is non-degenerate on a small ball around z_0 . On that ball $\omega_1 - \omega_0$ is closed, hence exact by the Poincaré lemma (Theorem 3.4): $\omega_1 - \omega_0 = d\sigma$, and σ can be chosen to vanish at z_0 by subtracting its constant part. Define a time-dependent vector field V_t by $\iota_{V_t}\omega_t = -\sigma$; it exists by non-degeneracy and vanishes at z_0 , so its flow ψ_t fixes z_0 and is defined near z_0 up to $t = 1$. The one tool used without proof is Cartan's formula for the rate of change of a form along a flow (da Silva, §6.1), which gives

$$\frac{d}{dt} \psi_t^* \omega_t = \psi_t^* (d \iota_{V_t} \omega_t + \iota_{V_t} d\omega_t + \omega_1 - \omega_0) = \psi_t^* (-d\sigma + 0 + d\sigma) = 0.$$

Closedness is spent in the middle term. So $\psi_1^* \omega_1 = \omega_0$, and the coordinates q, p composed with ψ_1^{-1} are a Darboux chart for ω . ■

The use [Lecture 5](#) and [Lecture 6](#) make of it

Theorem 4.20 (Darboux, adapted to commuting functions · *cited*). Let $F_1, \dots, F_n$ be functions with $\{F_i, F_j\} = 0$ and $dF_1, \dots, dF_n$ linearly independent at a point. Near that point there are functions $\psi_1, \dots, \psi_n$ such that (ψ, F) is a Darboux chart: $\omega = \sum_i d\psi_i \wedge dF_i$. (Bolsinov–Fomenko, Lemma 1.1, stated there in their convention $\omega = \sum dp \wedge dq$.)

In such a chart the F_i are momenta and the ψ_i positions, so by [Proposition 4.2](#) $X_{F_j} = \partial/\partial\psi_j$: the commuting flows are *translations* at unit speed. That is the whole of the Liouville–Arnold theorem *locally*, and it costs nothing — as Babelon, Bernard and Talon remark (ch. 2, Remark 2), near any point where $dH \neq 0$ Darboux gives canonical coordinates with $H = P_1$, so every Hamiltonian is “integrable” on a small enough patch.

So [Lecture 5](#) is not trivial. Its question is whether the ψ_i continue over a whole level set, and what they do when they are continued: they come back to themselves, which makes them angles and the level set a torus. Action-angle coordinates are an adapted Darboux chart on a whole neighbourhood of a torus, and the global part is what [Lecture 5](#) and [Lecture 6](#) add.

Reading

4.9 The Liouville volume $\omega^n/n!$, and more than volume

Proposition 4.21 (The Liouville volume form). In canonical coordinates,

$$\frac{\omega^n}{n!} = dq_1 \wedge dp_1 \wedge dq_2 \wedge dp_2 \wedge \cdots \wedge dq_n \wedge dp_n, \quad (4.12)$$

which as a measure is the phase-space volume of Liouville’s theorem (an exercise), $dq_1 \cdots dq_n dp_1 \cdots dp_n$. In particular ω^n is nowhere zero, on any symplectic manifold, since a Darboux chart exists around every point.

Proof. Expanding ω^n by bilinearity, a term with a repeated index repeats a 1-form and vanishes. The survivors are the $n!$ orderings of the 2-forms $dq_i \wedge dp_i$, which commute by (2.16) and so are all equal. Reordering into $dq_1 \wedge \cdots \wedge dq_n \wedge dp_1 \wedge \cdots \wedge dp_n$ costs a sign $(-1)^{n(n-1)/2}$, which a measure does not see. ■

Theorem 4.22 (Liouville, in one line · as [Proposition 4.8](#)). For any region R and any t for which φ_t is defined on all of R , $\text{vol}(\varphi_t(R)) = \text{vol}(R)$, where vol is the measure of $\omega^n/n!$.

Proof. By [Proposition 4.8](#) and the rule $F^*(\alpha \wedge \beta) = F^*\alpha \wedge F^*\beta$ (3.6), $\varphi_t^*(\omega^n/n!) = (\varphi_t^*\omega)^n/n! = \omega^n/n!$. Now change variables in the integral, φ_t being an orientation-preserving diffeomorphism onto its image: $\int_{\varphi_t(R)} \omega^n/n! = \int_R \varphi_t^*(\omega^n/n!) = \int_R \omega^n/n!$. ■

This is Liouville’s theorem on phase-space volume (an exercise), now a consequence of the stronger $\varphi_t^*\omega = \omega$. For $n = 1$ the two coincide; for $n \geq 2$ the symplectic statement is strictly stronger.

Theorem 4.23 (Poincaré’s integral invariant · *proved modulo Stokes*). Let γ be a closed loop in $\mathbb{R}^{2n}$ and let φ_t be defined on it. Then

$$\oint_{\varphi_t \circ \gamma} p \cdot dq = \oint_{\gamma} p \cdot dq.$$

The proof, Stokes on the tube the loop sweeps out (Figure 4.7), is in the reading.

Proof. Sweep the loop along the flow: $\Phi(s, u) = \varphi_s(\gamma(u))$ maps the cylinder $C = [0, t] \times S^1$ into phase space, with $\partial_s \Phi = X_H$. Orient C by $ds \wedge du$; its induced boundary is then the top loop minus the bottom one, so Stokes (Theorem 3.11) applied on C , as in Corollary 3.13, gives

$$\oint_{\varphi_t \circ \gamma} p \cdot dq - \oint_{\gamma} p \cdot dq = \int_C \Phi^* d(p \cdot dq) = - \int_C \Phi^* \omega,$$

using $d(p \cdot dq) = \sum dp_i \wedge dq_i = -\omega$. Relative to $ds \wedge du$ the integrand $\Phi^* \omega$ has coefficient $\omega(\partial_s \Phi, \partial_u \Phi) = \omega(X_H, \partial_u \Phi) = dH(\partial_u \Phi) = \partial_u(H \circ \Phi)$, by (4.1). At each fixed s this is the derivative of a periodic function of u , so its integral over $u \in S^1$ vanishes; by Fubini, so does $\int_C \Phi^* \omega$. ■

The same holds on any cotangent bundle, since the proof used only two facts that hold there globally: $d(p \cdot dq) = -\omega$ (Theorem 4.17) and $\iota_{X_H} \omega = dH$ (Definition 4.1).

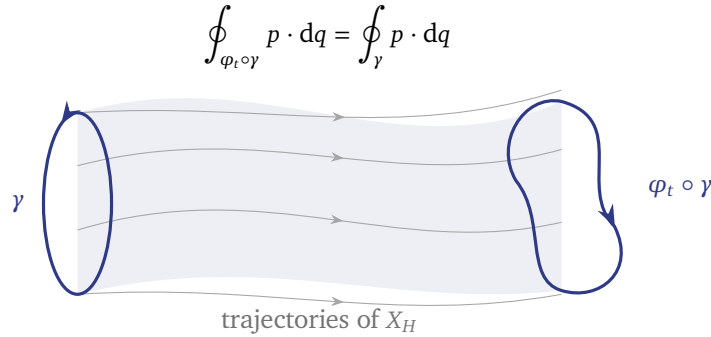


Figure 4.7: Poincaré's integral invariant: the loop is carried along the trajectories and deformed, and $\oint p \cdot dq$ does not change. By Stokes the change is $-\int \omega$ over the tube, and one edge of every cell of the tube is X_H , with $\omega(X_H, \cdot) = dH$, which integrates to zero around each loop.

For $n = 1$ this is Liouville again; for $n \geq 2$ it conserves the *sum of the signed areas of the loop's n shadows*, counted with winding number (a figure-eight shadow contributes the difference of its lobes), not a volume. It is the prototype of the action variable: Lecture 6 integrates $p \cdot dq$ around a cycle on an invariant torus, and the flow carries the cycle to another with the same integral.

Theorem 4.24 (Gromov non-squeezing · cited). *A symplectomorphism can map a ball of radius r in $\mathbb{R}^{2n}$ into the cylinder $\{q_1^2 + p_1^2 < R^2\}$ only if $r \leq R$. (Gromov, Invent. Math. **82**, 1985.)*

For $n \geq 2$ the cylinder has infinite volume, so this is symplectic, not volumetric — and not local, since it concerns a ball of definite size. We shall not use it.

The cross-section must be a (q_1, p_1) -plane. Over the (q_1, q_2) -plane the symplectic map $(q, p) \mapsto (\varepsilon q, p/\varepsilon)$ squeezes a ball of radius r into $\{q_1^2 + q_2^2 < \varepsilon^2 r^2\}$, as thin as you like.

Summary.

- *One structure.* $\iota_{X_f} \omega = df$ recovers Lecture 2's X_f , with $\omega(X_f, X_g) = \{f, g\}$ (Proposition 4.2), and the bracket of ω satisfies Jacobi because $d\omega = 0$.

- *Canonical transformations* are the maps with $F^*\omega = \omega$, and generating functions produce them because $d \circ d = 0$ ([Proposition 4.9](#)).
- *Lagrangian submanifolds*: dimension n , $\omega|_L = 0$. At most n independent functions commute ([Proposition 4.11](#)); graphs of closed 1-forms and the oscillator's tori are Lagrangian ([Proposition 4.14](#), [Example 4.15](#)).
- *Cotangent bundles* are symplectic with no choice made, and $\omega = -d(p \cdot dq)$ ([Theorem 4.17](#)).
- *Next*: [Lecture 5](#) proves the Liouville–Arnold theorem. Where n integrals in involution are independent, the compact components of their common level sets are tori, and the motion on them is a straight line.

Tutorial. Exercise Sheet 4.

References

da Silva, *Lectures on Symplectic Geometry*, §§1–4 (symplectic forms, cotangent bundles and the tautological form, Lagrangian submanifolds, generating functions) and §§7–8 (the Moser trick and Darboux); Arnold, *Mathematical Methods of Classical Mechanics*, §§37–38 and 40–41 (symplectic structures, integral invariants, the Poisson bracket from ω), §43 (Darboux), §44 (the integral invariant of Poincaré–Cartan) and §§47–48 (Hamilton–Jacobi, generating functions); Bolsinov and Fomenko, *Integrable Hamiltonian Systems*, §1.3 (Darboux, and its version adapted to commuting functions). **Conventions.** da Silva's agree with ours exactly — $\omega = -d\alpha$ with α the tautological form, $\iota_{X_H}\omega = dH$, $\{f, g\} = \omega(X_f, X_g)$, and the engine's minus sign. Arnold and Bolsinov–Fomenko write $\omega = \sum dp \wedge dq$, so their ω is minus ours: check any formula from them against [\(3.13\)](#) and [Theorem 4.17](#) before transcribing it.